NIKUNJ BHAVSAR

MSc Data Science · University of Bristol
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PROFILE

MSc Data Science student at the University of Bristol with strong foundations in machine learning, deep learning, and scalable data engineering. Proven expertise in designing and deploying end-to-end ML pipelines using Python, TensorFlow, and enterprise cloud platforms. Strong analytical problem-solver, eager to leverage advanced predictive modeling and cloud architectures to drive data-centric business decisions and optimize operational workflows.

SKILLS

Languages: Python (primary), Java, R, JavaScript, C, HTML/CSS
ML & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, CNNs, NLP, Prompt Engineering
Data: SQL, PostgreSQL, MySQL, Pipeline Creation, ETL, API Data Extraction
Cloud & DevOps: AWS (Cloud Foundations certified), Git, CI/CD, Automation & Scripting
Visualisation: Power BI, Tableau, React, NodeJS
Other: Data Wrangling, Statistical Modelling, OOP, Agile, Responsible / Ethical AI

PROJECTS

Indian Sign Language Translator (Python, TensorFlow, ResNet50, OpenCV)

- Fine-tuned a ResNet50 deep learning model on ~51,000 ISL alphabet images, achieving 96.4% validation accuracy -demonstrating proficiency in transfer learning and model optimisation.
- Engineered a real-time prediction pipeline using OpenCV to process live webcam feeds, translating sign gestures to text instantly; showcases end-to-end ML deployment skills relevant to production AI systems.
- Applied data augmentation and regularisation techniques to improve generalisation -key skills for bias mitigation and responsible AI.

UK Housing Investment Analysis (Python, Pandas, Data Visualisation)

- Conducted a comprehensive data science analysis of the UK housing market to surface investment insights, used the model to propose location for data centre at ACC 2026, saving Accuracy $50M.
- Applied statistical methods and data visualisation to identify regional price trends, affordability patterns, and market signals, translating raw data into actionable, business-relevant conclusions.
- Took data from the UK Government sites like police.uk, ons, etc.

Content-based Music Recommendation Engine (Python, KNN, TF-IDF)

- Architected an end-to-end machine learning pipeline to analyze and recommend similar music tracks from a dataset of 10,000 songs.
- Developed a content-based filtering model utilizing Principal Component Analysis (PCA) and K-Nearest Neighbors (KNN) to calculate track similarity based on tempo, timbre, and description (TF-IDF).

PCA for Diabetic Diagnosis (Python, Scikit-learn, Dimensionality Reduction)

- Applied Principal Component Analysis (PCA) to a medical dataset to reduce dimensionality and improve diagnostic classification accuracy, demonstrating proficiency in unsupervised ML techniques and statistical modelling.
- Evaluated model performance with appropriate metrics and visualised variance explained -showcasing ability to communicate complex ML outputs clearly and responsibly.

Voyager Mobile App (React Native, Google Places API, JavaScript)

- Built a cross-platform travel planning app consuming real-time Google Places API data, demonstrating API integration and scalable front-end engineering.

- Designed an intuitive itinerary interface that improved user engagement time by 40% during testing - evidence of user-centred, data-informed design thinking.

Parcel Management Software (Java, MVC, Swing)

- Implemented a Java backend using MVC architecture capable of managing 1,000+ transactions with full logging, demonstrating large-scale OOP and software engineering best practices.
- Built a dynamically updating Swing GUI reflecting real-time state changes, showing experience in building robust, interactive systems.

EXPERIENCE

Web Developer (Part-Time) | Rephrain -UKRI Research Centre | Dec 2025 -Present

- Contributing to a UKRI-backed national research centre focused on Privacy, Harm Reduction and Adversarial Influence Online -directly relevant to responsible AI, data ethics, and online safety, all core themes in AI governance framework.
- Supporting web infrastructure for a multi-institution academic research environment, applying software engineering best practices in a production context alongside active researchers.

Web Developer Intern | The Place To Be Gallery | Jan 2024 -May 2024

- Executed SEO and analytics strategies across WordPress sites, increasing organic traffic by 15% through data-informed content and structural improvements.
- Developed responsive, accessible website designs, and maintained regular automated backups to minimise downtime - demonstrating reliability and production-quality thinking.

EDUCATION

MSc Data Science | University of Bristol | 2025 - 2026

- Relevant modules: Statistical Computing, Large Scale Data Engineering, Visual Analytics, Introduction to AI & Text Analysis, Data Science Methods & Practices, Technology Innovation Business & Society.

BSc Computer Science | University of Hertfordshire | 2022 - 2025

- First Class Honours | GPA: 3.88
- Relevant modules: Machine Learning & Neural Computing, Artificial Intelligence, Algorithms & Data Structures, Software Architecture, Mobile Computing, Data Modelling for Databases.

CERTIFICATIONS

- AWS Academy Graduate -Cloud Foundations (October 2025)
- IBM -Data Analysis with Python (October 2025)